

WASP-3b

R-0296

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_180716 · created 2026-07-17T19:45:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458315.8515 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.039
Transit depth (Rp/Rs)^2	0.0002 +/- 0.011 [%]
Orbital Inclination (inc)	87.47 +/- 2.89
Airmass coefficient 1 (a1)	1.331 +/- 0.037
Airmass coefficient 2 (a2)	-0.282 +/- 0.089
Scatter in the residuals of the lightcurve fit is	2.68 %
Best Comparison Star	#1 - [348, 112]
Optimal Aperture	4.02
Optimal Annulus	9.82
Transit Duration (day)	0.114 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.039 vs 0.10538 (deviation 2.1σ)
Duration fit vs published	0.114 d vs 0.1167 d (deviation 0.19σ)
Mid-time O–C	-0.3 min
Depth SNR	0.0
Residual scatter	2.68 %

Light curve

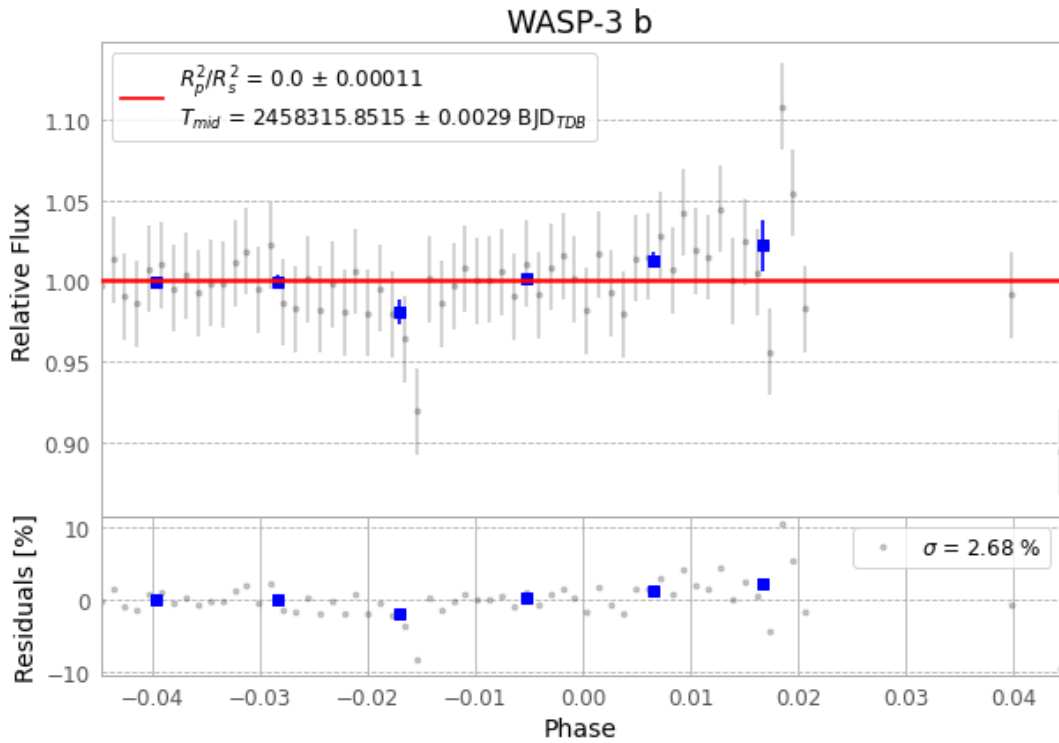


Figure 1 — FinalLightCurve_WASP-3 b_2018-07-15.png (SHA-256 f6dc64af8ca3b09c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef58d751db92b2f9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 96bb2ba

Data provenance

85 FITS frames (56 MB), WASP-3180716062112.FITS → WASP-3180716103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-180716.log (5746b23ae608...), FinalLightCurve_WASP-3 b_2018-07-15.png (f6dc64af8ca3...), FinalLightCurve_WASP-3 b_2018-07-15.csv (7b98b4869411...)

Compute resources

Wall time 190.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 19:45Z.